Management and Monitoring of Future Communication Networks: Exercise 01

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Exercise 1: Implementation of Random Access and Slotted ALOHA

- **Goal** of this exercise is **to identify pros and cons** using **slotted ALOHA** or **random access**/pure ALOHA. Both approaches are implemented in Python in the following
- **From theory, we know**
  - Random access has a maximal channel utilization of 18.4%
  - For slotted ALOHA, it is 36.8%

- **Exercise 1.1:** Implementation of random access and slotted ALOHA
  - Simulate the transmission of 100 – 1000 messages per hour in steps of 100 messages with random access and slotted ALOHA on a single transmission channel
  - The time to transmit a single message is 0.999s, the slot length for slotted ALOHA is 1s
  - All messages transmitted simultaneously are colliding
  - Calculate the number of collisions and the collision probability for all message counts per hour
  - **Note:** To achieve the collision probability, the pandas package Interval can help
  - What are key insights analyzing the collision probabilities?
  - **Note:** The collision probability p for random access with a total investigated time frame T can be achieved using the following equation to double check your simulation results
  - $p = 1 - (\prod_{a \in S}(1 - \frac{T_s + T_a}{T})$ considering transmissions s and a with times T_s and T_a

- **Exercise 1.2:** Change the time to transmit single messages to a random number between 0 and 2s and compare results to exercise 1.1 results. Test for slot lengths of 1s and 2s. What are key insights?
- **Bonus:** Repeat simulations for 1.1 and 1.2 several times. What can you do to improve result quality and statement?